

Multiple Choice

1. **Answer:** C

Options: A) Bubble sort; B) Insertion sort; C) Merge sort; D) Quicksort

Note: Correct option: C - Merge sort

2. **Answer:** D

Options: A) $O(\log N)$; B) $O(N \log N)$; C) $O(\log N) + O(1)$; D) $O((\log N)^2)$

Note: Correct option: D - $O((\log N)^2)$

3. **Answer:** B

Options: A) 3; B) 4; C) 2; D) 8

Note: Correct option: B - 4

4. **Answer:** D

Options: A) To translate Web addresses to host names; B) To determine the IP address of a given host name; C) To determine the hardware address of a given host name; D) To determine the hardware address of a given IP address

Note: Correct option: D - To determine the hardware address of a given IP address

5. **Answer:** A

Options: A) Replace the page whose next reference will be the longest time in the future.; B) Replace the page whose next reference will be the shortest time in the future.; C) Replace the page whose most recent reference was the shortest time in the past.; D) Replace the page whose most recent reference was the longest time in the past.

Note: Correct option: A - Replace the page whose next reference will be the longest time in the future.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'The program may have bugs.'

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
10. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** `def get__odd__occurence(arr, arr_size): | return arr[i] | return -1`
Note: Students should provide a detailed explanation referencing algorithm steps.
12. **Answer:** `def sort_sublists(input_list): | return result`
Note: Students should provide a detailed explanation referencing algorithm steps.

End of Answer Key